

Generation Interconnection System Impact Study Report

For Queue Project Q13-0239

May Street 138 kV

April 30, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0239. The Interconnection Customer proposes a 241.5 MW wind facility interconnecting at the May Street 138 kV substation in the Kentucky area. The request was received on May 20, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 146 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,926,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Juliette to Alibates 138 kV is overloaded to 106.8% of its 241 MVA emergency rating for the loss of the Carraizo 1201 to Pine Bluff Watson Chapel 345 kV L. The project contributes 33.0% DFAX (79.7 MW) to this overload.

The Action to Eldorado 138 kV is overloaded to 100.5% of its 475 MVA emergency rating for the loss of the Carraizo 1201 to Pine Bluff Watson Chapel 345 kV. The project contributes 35.2% DFAX (85.1 MW) to this overload.

Action to Alibates 138 kV: 100.1% loading (510 MVA) following the outage of the Pine Bluff Watson Chapel Substation to Carraizo 1201 345 kV Line; DFAX = 35.2%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Juliette to Alibates 138 kV	Reconductor line	\$4,475,000	70% (\$3,122,000)
Action to Eldorado 138 kV	Reconductor line	\$1,061,000	51% (\$544,000)
Alibates to Action 138 kV	Reconductor line	\$8,306,000	51% (\$4,260,000)
	Total allocated to project		\$7,926,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.